Additional file 2

What Does EEG-to-fMRI Synthesis Add to EEG-Based Diagnosis? A Controlled Benchmark in Psychosis, Schizophrenia and Parkinson's Disease

Supplementary tables

Table S1: Added value of synthesized fMRI, full precision. ΔAUC of the combined view over the baseline; 95% and Bonferroni-level (40 comparisons) percentile intervals of a paired stratified bootstrap over subjects (20,000 resamples); two-sided bootstrap p; Holm-adjusted p within the 20 trained-synthesizer comparisons (XP1: within the 20 XP1 comparisons) and within the 40 comparisons including untrained synthesizers.

Baseline	Cohort	Synthesizer	Weights	ΔAUC	95% CI	Bonferroni-40 CI	p	Holm (20)	Holm (40)
band power	FEP1	CNN-TAG	trained (NODDI)	−0.042	[−0.113, +0.029]	[−0.159, +0.076]	0.2535	1.000	1.000
band power	FEP1	CNN-TAG	untrained	−0.095	[−0.163, −0.036]	[−0.220, −0.004]	0.0005	–	0.019
band power	FEP1	CNN-TAG	trained (XP1)	−0.130	[−0.223, −0.043]	[−0.285, +0.017]	0.0035	0.063	–
band power	FEP1	E2fNet	trained (NODDI)	+0.006	[−0.074, +0.081]	[−0.123, +0.136]	0.8799	1.000	1.000
band power	FEP1	E2fNet	untrained	−0.013	[−0.071, +0.046]	[−0.112, +0.093]	0.6469	–	1.000
band power	FEP1	E2fNet	trained (XP1)	−0.043	[−0.125, +0.036]	[−0.188, +0.082]	0.2989	1.000	–
band power	FEP1	Spec2VolCAMU-Net	trained (NODDI)	−0.052	[−0.123, +0.014]	[−0.175, +0.060]	0.1370	1.000	1.000
band power	FEP1	Spec2VolCAMU-Net	untrained	+0.006	[−0.051, +0.062]	[−0.092, +0.100]	0.8392	–	1.000
band power	FEP1	Spec2VolCAMU-Net	trained (XP1)	−0.105	[−0.177, −0.038]	[−0.234, +0.001]	0.0014	0.027	–
band power	FEP1	NT-ViT	trained (NODDI)	−0.028	[−0.091, +0.031]	[−0.140, +0.065]	0.3525	1.000	1.000
band power	FEP1	NT-ViT	untrained	+0.018	[−0.035, +0.072]	[−0.073, +0.107]	0.5014	–	1.000
band power	FEP1	NT-ViT	trained (XP1)	+0.013	[−0.045, +0.070]	[−0.088, +0.107]	0.6463	1.000	–
band power	FEP2	CNN-TAG	trained (NODDI)	−0.024	[−0.068, +0.017]	[−0.104, +0.047]	0.2648	1.000	1.000
band power	FEP2	CNN-TAG	untrained	+0.028	[−0.016, +0.083]	[−0.042, +0.133]	0.2368	–	1.000
band power	FEP2	CNN-TAG	trained (XP1)	+0.025	[−0.031, +0.086]	[−0.073, +0.134]	0.3916	1.000	–
band power	FEP2	E2fNet	trained (NODDI)	−0.003	[−0.042, +0.037]	[−0.069, +0.067]	0.8823	1.000	1.000
band power	FEP2	E2fNet	untrained	+0.001	[−0.037, +0.038]	[−0.066, +0.065]	0.9637	–	1.000
band power	FEP2	E2fNet	trained (XP1)	−0.004	[−0.042, +0.032]	[−0.068, +0.062]	0.8328	1.000	–
band power	FEP2	Spec2VolCAMU-Net	trained (NODDI)	+0.023	[−0.026, +0.072]	[−0.061, +0.109]	0.3550	1.000	1.000
band power	FEP2	Spec2VolCAMU-Net	untrained	+0.005	[−0.037, +0.046]	[−0.068, +0.074]	0.8046	–	1.000
band power	FEP2	Spec2VolCAMU-Net	trained (XP1)	−0.032	[−0.077, +0.009]	[−0.112, +0.038]	0.1250	1.000	–
band power	FEP2	NT-ViT	trained (NODDI)	+0.012	[−0.020, +0.045]	[−0.043, +0.074]	0.4683	1.000	1.000
band power	FEP2	NT-ViT	untrained	+0.001	[−0.034, +0.036]	[−0.061, +0.062]	0.9647	–	1.000
band power	FEP2	NT-ViT	trained (XP1)	−0.000	[−0.022, +0.023]	[−0.039, +0.039]	1.0000	1.000	–
band power	Semmelweis	CNN-TAG	trained (NODDI)	+0.007	[−0.053, +0.068]	[−0.093, +0.111]	0.8459	1.000	1.000
band power	Semmelweis	CNN-TAG	untrained	+0.073	[+0.001, +0.150]	[−0.047, +0.207]	0.0477	–	1.000

Baseline	Cohort	Synthesizer	Weights	ΔAUC	95% CI	Bonferroni-40 CI	p	Holm (20)	Holm (40)
band power	Semmelweis	CNN-TAG	trained (XP1)	+0.032	[−0.049, +0.116]	[−0.104, +0.175]	0.4455	1.000	–
band power	Semmelweis	E2fNet	trained (NODDI)	+0.067	[+0.013, +0.125]	[−0.022, +0.166]	0.0158	0.269	0.521
band power	Semmelweis	E2fNet	untrained	−0.001	[−0.061, +0.059]	[−0.103, +0.098]	0.9866	–	1.000
band power	Semmelweis	E2fNet	trained (XP1)	+0.070	[+0.008, +0.136]	[−0.039, +0.185]	0.0291	0.466	–
band power	Semmelweis	Spec2VolCAMU-Net	trained (NODDI)	+0.112	[+0.035, +0.196]	[−0.009, +0.261]	0.0038	0.072	0.133
band power	Semmelweis	Spec2VolCAMU-Net	untrained	+0.165	[+0.089, +0.248]	[+0.046, +0.319]	0.0000	–	0.000
band power	Semmelweis	Spec2VolCAMU-Net	trained (XP1)	+0.155	[+0.078, +0.241]	[+0.036, +0.309]	0.0000	0.000	–
band power	Semmelweis	NT-ViT	trained (NODDI)	−0.013	[−0.107, +0.086]	[−0.163, +0.160]	0.7660	1.000	1.000
band power	Semmelweis	NT-ViT	untrained	+0.096	[+0.032, +0.166]	[−0.010, +0.223]	0.0034	–	0.122
band power	Semmelweis	NT-ViT	trained (XP1)	−0.039	[−0.121, +0.042]	[−0.182, +0.095]	0.3468	1.000	–
band power	PD	CNN-TAG	trained (NODDI)	+0.019	[−0.006, +0.053]	[−0.021, +0.086]	0.1623	1.000	1.000
band power	PD	CNN-TAG	untrained	+0.013	[−0.026, +0.050]	[−0.065, +0.082]	0.4672	–	1.000
band power	PD	CNN-TAG	trained (XP1)	−0.018	[−0.054, +0.013]	[−0.093, +0.034]	0.2712	1.000	–
band power	PD	E2fNet	trained (NODDI)	+0.005	[−0.024, +0.038]	[−0.045, +0.073]	0.8016	1.000	1.000
band power	PD	E2fNet	untrained	+0.029	[−0.002, +0.066]	[−0.021, +0.101]	0.0644	–	1.000
band power	PD	E2fNet	trained (XP1)	−0.008	[−0.051, +0.030]	[−0.092, +0.065]	0.7287	1.000	–
band power	PD	Spec2VolCAMU-Net	trained (NODDI)	+0.035	[+0.010, +0.072]	[+0.000, +0.107]	0.0025	0.050	0.095
band power	PD	Spec2VolCAMU-Net	untrained	−0.019	[−0.077, +0.026]	[−0.130, +0.056]	0.4917	–	1.000
band power	PD	Spec2VolCAMU-Net	trained (XP1)	−0.003	[−0.061, +0.040]	[−0.112, +0.074]	1.0000	1.000	–
band power	PD	NT-ViT	trained (NODDI)	+0.038	[+0.010, +0.078]	[−0.010, +0.117]	0.0092	0.166	0.313
band power	PD	NT-ViT	untrained	+0.027	[−0.000, +0.062]	[−0.023, +0.094]	0.0624	–	1.000
band power	PD	NT-ViT	trained (XP1)	+0.021	[−0.003, +0.051]	[−0.022, +0.083]	0.1098	1.000	–
band power	Fribourg	CNN-TAG	trained (NODDI)	−0.000	[−0.072, +0.068]	[−0.129, +0.123]	1.0000	1.000	1.000
band power	Fribourg	CNN-TAG	untrained	+0.002	[−0.066, +0.075]	[−0.121, +0.136]	0.9677	–	1.000
band power	Fribourg	CNN-TAG	trained (XP1)	−0.050	[−0.136, +0.031]	[−0.198, +0.083]	0.2366	1.000	–
band power	Fribourg	E2fNet	trained (NODDI)	+0.079	[+0.004, +0.160]	[−0.044, +0.228]	0.0352	0.563	1.000
band power	Fribourg	E2fNet	untrained	+0.061	[+0.018, +0.118]	[−0.004, +0.169]	0.0026	–	0.096
band power	Fribourg	E2fNet	trained (XP1)	+0.077	[+0.020, +0.151]	[−0.011, +0.213]	0.0084	0.143	–
band power	Fribourg	Spec2VolCAMU-Net	trained (NODDI)	+0.013	[−0.048, +0.066]	[−0.101, +0.105]	0.6147	1.000	1.000
band power	Fribourg	Spec2VolCAMU-Net	untrained	+0.042	[−0.009, +0.105]	[−0.044, +0.160]	0.1232	–	1.000
band power	Fribourg	Spec2VolCAMU-Net	trained (XP1)	+0.011	[−0.061, +0.081]	[−0.122, +0.128]	0.7604	1.000	–
band power	Fribourg	NT-ViT	trained (NODDI)	−0.007	[−0.048, +0.031]	[−0.089, +0.058]	0.7827	1.000	1.000
band power	Fribourg	NT-ViT	untrained	+0.033	[−0.009, +0.081]	[−0.042, +0.126]	0.1349	–	1.000
band power	Fribourg	NT-ViT	trained (XP1)	+0.018	[−0.022, +0.057]	[−0.063, +0.092]	0.3569	1.000	–
information-matched EEG	FEP1	CNN-TAG	trained (NODDI)	−0.008	[−0.132, +0.118]	[−0.225, +0.204]	0.9143	1.000	1.000
information-matched EEG	FEP1	CNN-TAG	untrained	−0.133	[−0.243, −0.031]	[−0.315, +0.031]	0.0099	–	0.376
information-matched EEG	FEP1	CNN-TAG	trained (XP1)	−0.030	[−0.160, +0.093]	[−0.252, +0.167]	0.6458	1.000	–
information-matched EEG	FEP1	E2fNet	trained (NODDI)	−0.070	[−0.183, +0.038]	[−0.260, +0.103]	0.2139	1.000	1.000
information-matched EEG	FEP1	E2fNet	untrained	+0.018	[−0.079, +0.113]	[−0.140, +0.163]	0.6829	–	1.000
information-matched EEG	FEP1	E2fNet	trained (XP1)	−0.045	[−0.162, +0.067]	[−0.239, +0.141]	0.4408	1.000	–
information-matched EEG	FEP1	Spec2VolCAMU-Net	trained (NODDI)	+0.018	[−0.107, +0.138]	[−0.195, +0.207]	0.7698	1.000	1.000
information-matched EEG	FEP1	Spec2VolCAMU-Net	untrained	−0.036	[−0.116, +0.040]	[−0.167, +0.089]	0.3600	–	1.000
information-matched EEG	FEP1	Spec2VolCAMU-Net	trained (XP1)	−0.101	[−0.212, +0.006]	[−0.296, +0.069]	0.0659	1.000	–
information-matched EEG	FEP1	NT-ViT	trained (NODDI)	−0.044	[−0.126, +0.038]	[−0.191, +0.091]	0.3011	1.000	1.000
information-matched EEG	FEP1	NT-ViT	untrained	+0.055	[−0.016, +0.132]	[−0.055, +0.191]	0.1348	–	1.000
information-matched EEG	FEP1	NT-ViT	trained (XP1)	+0.043	[−0.028, +0.118]	[−0.072, +0.166]	0.2335	1.000	–
information-matched EEG	FEP2	CNN-TAG	trained (NODDI)	−0.226	[−0.354, −0.099]	[−0.449, −0.012]	0.0005	0.010	0.020
information-matched EEG	FEP2	CNN-TAG	untrained	−0.080	[−0.200, +0.034]	[−0.281, +0.109]	0.1825	–	1.000
information-matched EEG	FEP2	CNN-TAG	trained (XP1)	+0.033	[−0.108, +0.177]	[−0.196, +0.275]	0.6409	1.000	–

Baseline	Cohort	Synthesizer	Weights	ΔAUC	95% CI	Bonferroni-40 CI	p	Holm (20)	Holm (40)
information-matched EEG	FEP2	E2fNet	trained (NODDI)	−0.033	[−0.124, +0.052]	[−0.186, +0.105]	0.4662	1.000	1.000
information-matched EEG	FEP2	E2fNet	untrained	−0.037	[−0.137, +0.055]	[−0.209, +0.112]	0.4562	–	1.000
information-matched EEG	FEP2	E2fNet	trained (XP1)	−0.063	[−0.176, +0.048]	[−0.253, +0.120]	0.2724	1.000	–
information-matched EEG	FEP2	Spec2VolCAMU-Net	trained (NODDI)	+0.057	[−0.070, +0.183]	[−0.158, +0.267]	0.3763	1.000	1.000
information-matched EEG	FEP2	Spec2VolCAMU-Net	untrained	−0.017	[−0.094, +0.055]	[−0.148, +0.106]	0.6594	–	1.000
information-matched EEG	FEP2	Spec2VolCAMU-Net	trained (XP1)	−0.075	[−0.191, +0.035]	[−0.275, +0.110]	0.1836	1.000	–
information-matched EEG	FEP2	NT-ViT	trained (NODDI)	−0.015	[−0.086, +0.056]	[−0.134, +0.103]	0.6718	1.000	1.000
information-matched EEG	FEP2	NT-ViT	untrained	−0.009	[−0.095, +0.073]	[−0.158, +0.126]	0.8660	–	1.000
information-matched EEG	FEP2	NT-ViT	trained (XP1)	−0.011	[−0.089, +0.066]	[−0.144, +0.116]	0.7961	1.000	–
information-matched EEG	Semmelweis	CNN-TAG	trained (NODDI)	+0.093	[−0.034, +0.221]	[−0.113, +0.310]	0.1532	1.000	1.000
information-matched EEG	Semmelweis	CNN-TAG	untrained	+0.072	[−0.040, +0.186]	[−0.116, +0.263]	0.2149	–	1.000
information-matched EEG	Semmelweis	CNN-TAG	trained (XP1)	+0.004	[−0.115, +0.120]	[−0.195, +0.198]	0.9356	1.000	–
information-matched EEG	Semmelweis	E2fNet	trained (NODDI)	+0.078	[−0.033, +0.192]	[−0.107, +0.268]	0.1732	1.000	1.000
information-matched EEG	Semmelweis	E2fNet	untrained	+0.036	[−0.063, +0.133]	[−0.131, +0.193]	0.4822	–	1.000
information-matched EEG	Semmelweis	E2fNet	trained (XP1)	+0.036	[−0.061, +0.130]	[−0.130, +0.191]	0.4517	1.000	–
information-matched EEG	Semmelweis	Spec2VolCAMU-Net	trained (NODDI)	+0.004	[−0.096, +0.101]	[−0.159, +0.165]	0.9385	1.000	1.000
information-matched EEG	Semmelweis	Spec2VolCAMU-Net	untrained	+0.169	[+0.073, +0.265]	[+0.011, +0.327]	0.0006	–	0.023
information-matched EEG	Semmelweis	Spec2VolCAMU-Net	trained (XP1)	+0.085	[−0.014, +0.187]	[−0.078, +0.259]	0.0945	1.000	–
information-matched EEG	Semmelweis	NT-ViT	trained (NODDI)	−0.021	[−0.161, +0.117]	[−0.253, +0.210]	0.7733	1.000	1.000
information-matched EEG	Semmelweis	NT-ViT	untrained	+0.026	[−0.046, +0.099]	[−0.092, +0.150]	0.4827	–	1.000
information-matched EEG	Semmelweis	NT-ViT	trained (XP1)	−0.013	[−0.145, +0.122]	[−0.231, +0.204]	0.8468	1.000	–
information-matched EEG	PD	CNN-TAG	trained (NODDI)	−0.070	[−0.176, +0.034]	[−0.246, +0.096]	0.1924	1.000	1.000
information-matched EEG	PD	CNN-TAG	untrained	−0.027	[−0.141, +0.075]	[−0.228, +0.138]	0.6488	–	1.000
information-matched EEG	PD	CNN-TAG	trained (XP1)	−0.083	[−0.197, +0.024]	[−0.274, +0.103]	0.1308	1.000	–
information-matched EEG	PD	E2fNet	trained (NODDI)	−0.008	[−0.066, +0.054]	[−0.114, +0.102]	0.7752	1.000	1.000
information-matched EEG	PD	E2fNet	untrained	−0.056	[−0.122, −0.002]	[−0.181, +0.033]	0.0429	–	1.000
information-matched EEG	PD	E2fNet	trained (XP1)	−0.078	[−0.173, +0.005]	[−0.234, +0.059]	0.0653	1.000	–
information-matched EEG	PD	Spec2VolCAMU-Net	trained (NODDI)	+0.014	[−0.053, +0.086]	[−0.098, +0.142]	0.6928	1.000	1.000
information-matched EEG	PD	Spec2VolCAMU-Net	untrained	−0.014	[−0.109, +0.074]	[−0.196, +0.138]	0.7661	–	1.000
information-matched EEG	PD	Spec2VolCAMU-Net	trained (XP1)	−0.008	[−0.096, +0.090]	[−0.156, +0.170]	0.8534	1.000	–
information-matched EEG	PD	NT-ViT	trained (NODDI)	+0.006	[−0.051, +0.064]	[−0.095, +0.106]	0.8353	1.000	1.000
information-matched EEG	PD	NT-ViT	untrained	+0.024	[−0.024, +0.077]	[−0.070, +0.118]	0.3317	–	1.000
information-matched EEG	PD	NT-ViT	trained (XP1)	+0.000	[−0.056, +0.056]	[−0.102, +0.095]	1.0000	1.000	–
information-matched EEG	Fribourg	CNN-TAG	trained (NODDI)	+0.018	[−0.178, +0.215]	[−0.307, +0.327]	0.8447	1.000	1.000
information-matched EEG	Fribourg	CNN-TAG	untrained	−0.029	[−0.151, +0.094]	[−0.238, +0.183]	0.6619	–	1.000
information-matched EEG	Fribourg	CNN-TAG	trained (XP1)	−0.127	[−0.272, +0.022]	[−0.370, +0.136]	0.0932	1.000	–
information-matched EEG	Fribourg	E2fNet	trained (NODDI)	−0.029	[−0.160, +0.099]	[−0.261, +0.188]	0.6643	1.000	1.000
information-matched EEG	Fribourg	E2fNet	untrained	+0.004	[−0.121, +0.127]	[−0.209, +0.212]	0.9359	–	1.000
information-matched EEG	Fribourg	E2fNet	trained (XP1)	+0.035	[−0.105, +0.180]	[−0.193, +0.283]	0.6435	1.000	–
information-matched EEG	Fribourg	Spec2VolCAMU-Net	trained (NODDI)	+0.009	[−0.154, +0.162]	[−0.261, +0.263]	0.9010	1.000	1.000
information-matched EEG	Fribourg	Spec2VolCAMU-Net	untrained	−0.013	[−0.116, +0.096]	[−0.178, +0.181]	0.8070	–	1.000
information-matched EEG	Fribourg	Spec2VolCAMU-Net	trained (XP1)	−0.081	[−0.235, +0.061]	[−0.342, +0.156]	0.2788	1.000	–
information-matched EEG	Fribourg	NT-ViT	trained (NODDI)	+0.068	[−0.064, +0.202]	[−0.166, +0.295]	0.3006	1.000	1.000
information-matched EEG	Fribourg	NT-ViT	untrained	−0.018	[−0.158, +0.112]	[−0.254, +0.193]	0.8343	–	1.000
information-matched EEG	Fribourg	NT-ViT	trained (XP1)	−0.061	[−0.149, +0.020]	[−0.224, +0.076]	0.1493	1.000	–

Table S2: AUC of every view: mean over 20 repeats of stratified 10-fold cross-validation, 95% bootstrap interval over subjects and one-sided label-permutation p (1,000 permutations for EEG views, 500 for NODDI-trained and untrained synthesized views, 100 for XP1-trained and 50-epoch views; – not tested).

Cohort	View	AUC	95% CI	p
FEP1	Band power (relative)	0.620	[0.48, 0.75]	0.072
	Band power (absolute)	0.572	[0.44, 0.71]	0.210
	Band power, 1–30 Hz	0.732	[0.63, 0.85]	–
	Information-matched (spectra mean, SD)	0.522	[0.39, 0.66]	0.355
	Information-matched (random projection)	0.540	[0.41, 0.69]	–
	CNN-TAG, trained (NODDI)	0.573	[0.46, 0.72]	0.202
	CNN-TAG, untrained	0.413	[0.28, 0.54]	0.806
	CNN-TAG, trained (XP1)	0.495	[0.36, 0.64]	0.475
	E2fNet, trained (NODDI)	0.527	[0.40, 0.67]	0.327
	E2fNet, untrained	0.632	[0.50, 0.76]	0.060
	E2fNet, trained (XP1)	0.553	[0.42, 0.69]	0.188
	Spec2VolCAMU-Net, trained (NODDI)	0.599	[0.46, 0.74]	0.116
	Spec2VolCAMU-Net, untrained	0.516	[0.39, 0.65]	0.381
	Spec2VolCAMU-Net, trained (XP1)	0.508	[0.36, 0.63]	0.426
	NT-ViT, trained (NODDI)	0.496	[0.38, 0.63]	0.415
	NT-ViT, untrained	0.605	[0.49, 0.74]	0.080
	NT-ViT, trained (XP1)	0.627	[0.51, 0.77]	0.069
	CNN-TAG, 50 epochs	0.495	[0.36, 0.63]	0.515
FEP2	Band power (relative)	0.646	[0.51, 0.78]	0.050
	Band power (absolute)	0.573	[0.44, 0.72]	0.225
	Band power, 1–30 Hz	0.646	[0.52, 0.79]	–
	Information-matched (spectra mean, SD)	0.566	[0.43, 0.72]	0.248
	Information-matched (random projection)	0.604	[0.47, 0.77]	–
	CNN-TAG, trained (NODDI)	0.329	[0.18, 0.45]	0.960
	CNN-TAG, untrained	0.549	[0.42, 0.71]	0.238
	CNN-TAG, trained (XP1)	0.544	[0.40, 0.69]	0.366
	E2fNet, trained (NODDI)	0.574	[0.42, 0.72]	0.198
	E2fNet, untrained	0.565	[0.41, 0.71]	0.222
	E2fNet, trained (XP1)	0.487	[0.32, 0.61]	0.515
	Spec2VolCAMU-Net, trained (NODDI)	0.608	[0.47, 0.75]	0.118
	Spec2VolCAMU-Net, untrained	0.574	[0.44, 0.73]	0.202
	Spec2VolCAMU-Net, trained (XP1)	0.466	[0.29, 0.59]	0.653
	NT-ViT, trained (NODDI)	0.601	[0.47, 0.75]	0.114
	NT-ViT, untrained	0.626	[0.50, 0.78]	0.048
	NT-ViT, trained (XP1)	0.571	[0.43, 0.72]	0.218
	CNN-TAG, 50 epochs	0.484	[0.35, 0.64]	0.515
Semmelweis	Band power (relative)	0.571	[0.45, 0.72]	0.194
	Band power (absolute)	0.564	[0.46, 0.72]	0.218
	Band power, 1–30 Hz	0.541	[0.40, 0.66]	–
	Information-matched (spectra mean, SD)	0.541	[0.42, 0.68]	0.260
	Information-matched (random projection)	0.606	[0.50, 0.74]	–
	CNN-TAG, trained (NODDI)	0.585	[0.46, 0.71]	0.144

Cohort	View	AUC	95% CI	p
	CNN-TAG, untrained	0.558	[0.41, 0.68]	0.170
	CNN-TAG, trained (XP1)	0.511	[0.38, 0.65]	0.386
	E2fNet, trained (NODDI)	0.533	[0.41, 0.67]	0.277
	E2fNet, untrained	0.468	[0.34, 0.59]	0.587
	E2fNet, trained (XP1)	0.594	[0.48, 0.74]	0.089
	Spec2VolCAMU-Net, trained (NODDI)	0.619	[0.50, 0.75]	0.058
	Spec2VolCAMU-Net, untrained	0.740	[0.65, 0.86]	0.006
	Spec2VolCAMU-Net, trained (XP1)	0.660	[0.55, 0.79]	0.030
	NT-ViT, trained (NODDI)	0.472	[0.35, 0.62]	0.645
	NT-ViT, untrained	0.571	[0.48, 0.72]	0.124
	NT-ViT, trained (XP1)	0.477	[0.34, 0.60]	0.564
	CNN-TAG, 50 epochs	0.735	[0.64, 0.86]	0.010
PD	Band power (relative)	0.786	[0.63, 0.92]	0.002
	Band power (absolute)	0.772	[0.64, 0.91]	0.002
	Band power, 1–30 Hz	0.776	[0.62, 0.91]	–
	Information-matched (spectra mean, SD)	0.825	[0.72, 0.95]	0.001
	Information-matched (random projection)	0.823	[0.72, 0.94]	–
	CNN-TAG, trained (NODDI)	0.483	[0.33, 0.65]	0.463
	CNN-TAG, untrained	0.699	[0.60, 0.88]	0.024
	CNN-TAG, trained (XP1)	0.518	[0.36, 0.69]	0.376
	E2fNet, trained (NODDI)	0.744	[0.63, 0.88]	0.010
	E2fNet, untrained	0.707	[0.56, 0.86]	0.026
	E2fNet, trained (XP1)	0.660	[0.53, 0.83]	0.059
	Spec2VolCAMU-Net, trained (NODDI)	0.694	[0.57, 0.85]	0.040
	Spec2VolCAMU-Net, untrained	0.739	[0.62, 0.89]	0.008
	Spec2VolCAMU-Net, trained (XP1)	0.676	[0.55, 0.85]	0.050
	NT-ViT, trained (NODDI)	0.672	[0.52, 0.83]	0.024
	NT-ViT, untrained	0.694	[0.53, 0.84]	0.014
	NT-ViT, trained (XP1)	0.624	[0.46, 0.77]	0.030
	CNN-TAG, 50 epochs	0.647	[0.52, 0.81]	0.129
Fribourg	Band power (relative)	0.539	[0.39, 0.74]	0.361
	Band power (absolute)	0.641	[0.47, 0.82]	0.108
	Band power, 1–30 Hz	0.559	[0.39, 0.73]	–
	Information-matched (spectra mean, SD)	0.559	[0.38, 0.74]	0.284
	Information-matched (random projection)	0.524	[0.36, 0.70]	–
	CNN-TAG, trained (NODDI)	0.598	[0.45, 0.79]	0.162
	CNN-TAG, untrained	0.552	[0.40, 0.75]	0.283
	CNN-TAG, trained (XP1)	0.366	[0.19, 0.53]	0.842
	E2fNet, trained (NODDI)	0.538	[0.37, 0.71]	0.365
	E2fNet, untrained	0.544	[0.38, 0.74]	0.327
	E2fNet, trained (XP1)	0.535	[0.38, 0.71]	0.327
	Spec2VolCAMU-Net, trained (NODDI)	0.608	[0.44, 0.80]	0.150
	Spec2VolCAMU-Net, untrained	0.490	[0.33, 0.68]	0.493
	Spec2VolCAMU-Net, trained (XP1)	0.515	[0.33, 0.68]	0.416
	NT-ViT, trained (NODDI)	0.630	[0.47, 0.81]	0.066
	NT-ViT, untrained	0.531	[0.35, 0.71]	0.305
	NT-ViT, trained (XP1)	0.342	[0.11, 0.38]	0.921

Cohort	View	AUC	95% CI	p
	CNN-TAG, 50 epochs	0.664	[0.53, 0.85]	0.059

Table S3: Recording metadata and artefact measures by group: patient and control means, AUC of the measure for the label (Mann–Whitney) and two-sided p; categorical fields with Fisher or χ^2 tests. Artefact measures: fraction of pseudo-TRs exceeding 150 μV peak-to-peak or an EMG-like 30–45 Hz power, median peak-to-peak amplitude (μV), median log 30–45 Hz power, log RMS of the EOG channels and (PD) of the accelerometer, over the analysed crop.

Cohort	Measure	Patients	Controls	AUC	p
FEP1	age	22.49	21.60	0.53	0.6609
FEP1	duration	306.40	307.82	0.48	0.7883
FEP1	file mb	78.44	83.98	0.47	0.6468
FEP1	lowpass	1500.0001500000149: 0/1; 500.0: 47/29 (pt/hc)			0.3896
FEP1	sex	F: 14/12; M: 33/18 (pt/hc)			0.4595
FEP1	sfreq	1000.0: 47/29; 3000.0003000000297: 0/1 (pt/hc)			0.3896
FEP1	eog log rms (artefact)	-10.504	-10.695	0.51	0.8879
FEP1	frac rejected (artefact)	0.574	0.695	0.38	0.0891
FEP1	hf logpow 30 45 (artefact)	-17.320	-17.056	0.41	0.1696
FEP1	median p2p uV (artefact)	419.003	284.109	0.43	0.2938
FEP1	ocular ICA components removed	1.13	1.07		0.6210
FEP2	age	23.38	24.19	0.47	0.7346
FEP2	duration	314.29	307.15	0.65	0.0437
FEP2	file mb	221.08	220.43	0.60	0.1907
FEP2	lowpass	1500.0001500000149: 27/27; 500.0: 4/3 (pt/hc)			1.0000
FEP2	sex	F: 9/14; M: 22/16 (pt/hc)			0.1919
FEP2	sfreq	1000.0: 4/3; 3000.0003000000297: 27/27 (pt/hc)			1.0000
FEP2	eog log rms (artefact)	-10.980	-10.739	0.37	0.0796
FEP2	frac rejected (artefact)	0.764	0.906	0.37	0.0745
FEP2	hf logpow 30 45 (artefact)	-17.159	-16.705	0.40	0.1774
FEP2	median p2p uV (artefact)	400.192	521.909	0.31	0.0109
FEP2	ocular ICA components removed	0.94	1.17		0.1481
Semmelweis	age	33.92	32.69	0.54	0.5853
Semmelweis	duration	135.68	129.58	0.36	0.0369
Semmelweis	edu years	10: 2/0; 11: 1/0; 11.5: 1/0; 12: 3/4; 13: 3/4; 14: 4/3; 14.5: 1/1; 15: 1/2; 16: 7/6; 17: 3/6; 18: 9/13; 8: 1/0; 9: 1/0 (pt/hc)			0.7433
Semmelweis	file mb	37.45	35.76	0.36	0.0369
Semmelweis	hour	14.03	16.33	0.28	0.0007
Semmelweis	month	7.05	6.62	0.56	0.3935
Semmelweis	n annot	0.11	0.28	0.50	0.9463
Semmelweis	sex	f: 14/15; m: 24/24 (pt/hc)			1.0000
Semmelweis	year	2016: 9/5; 2017: 18/23; 2018: 11/11 (pt/hc)			0.4190
Semmelweis	eog log rms (artefact)	-10.088	-10.403	0.60	0.1382
Semmelweis	frac rejected (artefact)	0.693	0.450	0.69	0.0033
Semmelweis	hf logpow 30 45 (artefact)	-17.465	-17.954	0.63	0.0574
Semmelweis	median p2p uV (artefact)	527.290	396.886	0.67	0.0120

Cohort	Measure	Patients	Controls	AUC	p
Semmelweis	ocular ICA components removed	1.11	0.79		0.0670
PD	age	69.68	69.32	0.53	0.6976
PD	duration	594.21	601.28	0.51	0.8996
PD	file mb	79.62	80.57	0.51	0.8996
PD	n annot	317.80	315.64	0.54	0.5864
PD	sex	Female: 9/9; Male: 16/16 (pt/hc)			1.0000
PD	acc log rms (artefact)	-4.423	-4.223	0.41	0.2772
PD	eog log rms (artefact)	-5.041	-4.494	0.26	0.0036
PD	frac rejected (artefact)	0.657	0.783	0.36	0.0806
PD	hf logpow 30 45 (artefact)	-16.953	-16.624	0.42	0.3320
PD	median p2p uV (artefact)	255.542	425.297	0.26	0.0036
PD	ocular ICA components removed	1.32	1.32		0.9383
Fribourg	duration	509.05	527.00	0.25	0.0035
Fribourg	file mb	279.18	284.93	0.50	0.9704
Fribourg	hour	9.00	9.88	0.29	0.0147
Fribourg	month	6.89	7.00	0.51	0.9022
Fribourg	n annot	783.74	775.50	0.47	0.7879
Fribourg	n channels	132: 1/24; 134: 18/0 (pt/hc)			0.0000
Fribourg	year	2014: 3/15; 2015: 7/9; 2016: 9/0 (pt/hc)			0.0002
Fribourg	eog log rms (artefact)	-10.443	-10.329	0.46	0.6334
Fribourg	frac rejected (artefact)	0.784	0.806	0.48	0.7866
Fribourg	hf logpow 30 45 (artefact)	-17.311	-17.170	0.43	0.4411
Fribourg	median p2p uV (artefact)	212.456	232.647	0.45	0.5990
Fribourg	ocular ICA components removed	0.42	0.42		0.8747

Table S4: Contrastive sinusoid separation: AUC (repeated stratified 10-fold, 3 repeats), 95% bootstrap interval and permutation p (20 permutations; minimum 0.048).

Cohort	Re-initialised encoder	Pretrained encoder	Band power
FEP1	0.485 [0.39, 0.67] (0.57)	0.599 [0.48, 0.73] (0.19)	0.620
FEP2	0.476 [0.34, 0.63] (0.52)	0.433 [0.27, 0.54] (0.76)	0.646
Semmelweis	0.538 [0.46, 0.72] (0.33)	0.562 [0.45, 0.70] (0.24)	0.571
PD	0.701 [0.58, 0.86] (0.048)	0.660 [0.53, 0.82] (0.048)	0.786
Fribourg	0.559 [0.38, 0.74] (0.33)	0.602 [0.44, 0.77] (0.14)	0.539

Table S5: Fidelity of the final synthesizers used for the clinical cohorts on the held-out validation subjects (NODDI 49 and 50; XP1 sub-xp109 and sub-xp110, pooled), and the leave-one-subject-out NODDI evaluation (16 test subjects; mean SSIM, RMSE ± SD over subjects and mean voxelwise temporal correlation r) with trigger-aligned EEG, EEG paired from the file start and ECG only. Shift nulls and selected epochs are in Table S6.

Data	Held-out subjects	Model	SSIM	RMSE	r
NODDI	49	EEG-free template	0.716	0.0750	+0.0000

NODDI	49	CNN-TAG	0.686	0.0847	-0.0118
NODDI	49	E2fNet	0.706	0.0888	+0.0028
NODDI	49	Spec2VolCAMU-Net	0.717	0.0792	+0.0001
NODDI	49	NT-ViT	0.690	0.0780	-0.0001
NODDI	50	EEG-free template	0.692	0.0803	+0.0000
NODDI	50	CNN-TAG	0.638	0.0936	+0.0004
NODDI	50	E2fNet	0.713	0.0915	+0.0020
NODDI	50	Spec2VolCAMU-Net	0.720	0.0866	+0.0150
NODDI	50	NT-ViT	0.700	0.0780	+0.0004
XP1	sub-xp109, sub-xp110	EEG-free template	0.684	0.0963	–
XP1	sub-xp109, sub-xp110	CNN-TAG (epoch 0)	0.699	0.0949	+0.0722
XP1	sub-xp109, sub-xp110	E2fNet (epoch 11)	0.690	0.0956	+0.0380
XP1	sub-xp109, sub-xp110	NT-ViT (epoch 21)	0.662	0.0914	-0.0002
XP1	sub-xp109, sub-xp110	Spec2VolCAMU-Net (epoch 8)	0.680	0.0978	+0.0045
NODDI LOSO (aligned)	16 subjects	EEG-free template	0.675	0.0884±0.0133	+0.0000
NODDI LOSO (aligned)	16 subjects	Ridge (spectra)	0.631	0.1018±0.0135	-0.0012
NODDI LOSO (aligned)	16 subjects	CNN-TAG	0.655	0.0981±0.0128	+0.0012
NODDI LOSO (aligned)	16 subjects	E2fNet	0.655	0.1055±0.0154	-0.0029
NODDI LOSO (aligned)	16 subjects	Spec2VolCAMU-Net	0.679	0.0936±0.0148	-0.0003
NODDI LOSO (filestart)	16 subjects	EEG-free template	0.675	0.0884±0.0133	+0.0000
NODDI LOSO (filestart)	16 subjects	Ridge (spectra)	0.631	0.1015±0.0133	-0.0014
NODDI LOSO (filestart)	16 subjects	CNN-TAG	0.640	0.1004±0.0142	-0.0040
NODDI LOSO (filestart)	16 subjects	E2fNet	0.661	0.1040±0.0162	+0.0015
NODDI LOSO (ecg only)	16 subjects	EEG-free template	0.675	0.0884±0.0133	+0.0000
NODDI LOSO (ecg only)	16 subjects	Ridge (spectra)	0.600	0.1087±0.0171	-0.0002
NODDI LOSO (ecg only)	16 subjects	CNN-TAG	0.650	0.0999±0.0132	-0.0039
NODDI LOSO (ecg only)	16 subjects	E2fNet	0.648	0.1094±0.0204	-0.0006

Table S6: Fidelity on NODDI, leave-one-subject-out (16 test subjects; mean ± SD over subjects, seeds averaged within a subject). PSNR from the RMSE of $[0, 1]$-scaled volumes (RMSE in Table S5). r: mean voxelwise temporal correlation with the real volumes; shift null: the same with circularly shifted EEG. Epoch: median selected epoch.

Input	Model	SSIM	PSNR	r	r (shift null)	Epoch
Aligned EEG	EEG-free template	0.675 ± 0.044	21.2 ± 1.2	$+0.000$	–	–
	Ridge (spectra)	0.631 ± 0.046	19.9 ± 1.2	-0.001	$+0.000$	–
	CNN-TAG	0.655 ± 0.048	20.2 ± 1.0	$+0.001$	-0.002	0
	E2fNet	0.655 ± 0.053	19.6 ± 1.1	-0.003	$+0.001$	10
	Spec2VolCAMU-Net	0.679 ± 0.052	20.7 ± 1.2	-0.000	$+0.000$	13
File-start EEG	Ridge (spectra)	0.631 ± 0.046	19.9 ± 1.1	-0.001	-0.000	–
	CNN-TAG	0.640 ± 0.047	20.0 ± 1.2	-0.004	$+0.000$	0
	E2fNet	0.661 ± 0.057	19.8 ± 1.2	$+0.002$	-0.000	10
ECG only	Ridge (spectra)	0.600 ± 0.060	19.4 ± 1.4	-0.000	-0.001	–
	CNN-TAG	0.650 ± 0.046	20.1 ± 1.1	-0.004	$+0.001$	0
	E2fNet	0.648 ± 0.056	19.4 ± 1.5	-0.001	$+0.000$	10

Table S7: Sensitivity analyses: AUC of each synthesized view with 20 (primary), 5, 10 or 40 principal components, average pooling in $3 \times 4 \times 4$-voxel blocks, covariates regressed out within training folds, ICA-cleaned inputs (ocular components removed), and synthesizers trained on XP1 (last column; for the last row of each cohort, CNN-TAG trained for 50 epochs). Covariates: age and sex (all cohorts with these data), chlorpromazine-equivalent dose (FEP1, FEP2), recording hour (Semmelweis, Fribourg), recording year (Fribourg).

Cohort	Synthesizer	Weights	$k = 20$	$k = 5$	$k = 10$	$k = 40$	Blocks	Covariates	ICA	XP1
FEP1	CNN-TAG	trained	0.57	0.55	0.49	0.53	0.58	0.60	0.50	0.50
		untrained	0.41	0.47	0.46	0.46	0.39	0.53	0.51	
	E2fNet	trained	0.53	0.58	0.58	0.51	0.50	0.52	0.58	0.55
		untrained	0.63	0.64	0.69	0.59	0.64	0.60	0.62	
	Spec2VolCAMU-Net	trained	0.60	0.47	0.46	0.65	0.57	0.57	0.62	0.51
		untrained	0.52	0.59	0.55	0.51	0.52	0.46	0.51	
	NT-ViT	trained	0.50	0.51	0.52	0.53	0.49	0.48	0.55	0.63
		untrained	0.60	0.55	0.60	0.54	0.58	0.48	0.41	
	Band power / information-matched		0.62 / 0.52					0.54 / 0.52	0.54 / 0.55	
	Band power without 30–45 Hz; CNN-TAG 50 epochs		0.73							0.49
FEP2	CNN-TAG	trained	0.33	0.42	0.35	0.41	0.36	0.44	0.43	0.54
		untrained	0.55	0.50	0.55	0.54	0.61	0.59	0.53	
	E2fNet	trained	0.57	0.45	0.56	0.52	0.55	0.52	0.54	0.49
		untrained	0.57	0.59	0.62	0.51	0.54	0.47	0.47	
	Spec2VolCAMU-Net	trained	0.61	0.58	0.53	0.62	0.59	0.48	0.32	0.47
		untrained	0.57	0.69	0.64	0.54	0.55	0.46	0.47	
	NT-ViT	trained	0.60	0.54	0.59	0.59	0.54	0.59	0.51	0.57
		untrained	0.63	0.62	0.61	0.62	0.55	0.52	0.49	
	Band power / information-matched		0.65 / 0.57					0.64 / 0.55	0.58 / 0.54	
	Band power without 30–45 Hz; CNN-TAG 50 epochs		0.65							0.48
Semmelweis	CNN-TAG	trained	0.58	0.53	0.60	0.63	0.61	0.54	0.47	0.51
		untrained	0.56	0.54	0.55	0.62	0.58	0.64	0.58	
	E2fNet	trained	0.53	0.49	0.46	0.66	0.53	0.50	0.57	0.59
		untrained	0.47	0.51	0.45	0.53	0.47	0.43	0.61	
	Spec2VolCAMU-Net	trained	0.62	0.68	0.67	0.62	0.64	0.63	0.63	0.66
		untrained	0.74	0.56	0.66	0.79	0.76	0.62	0.57	
	NT-ViT	trained	0.47	0.44	0.46	0.51	0.51	0.49	0.54	0.48
		untrained	0.57	0.56	0.54	0.57	0.57	0.54	0.61	
	Band power / information-matched		0.57 / 0.54					0.55 / 0.53	0.58 / 0.49	
	Band power without 30–45 Hz; CNN-TAG 50 epochs		0.54							0.74
PD	CNN-TAG	trained	0.48	0.62	0.58	0.48	0.47	0.50	0.68	0.52
		untrained	0.70	0.66	0.68	0.73	0.78	0.69	0.75	
	E2fNet	trained	0.74	0.67	0.66	0.74	0.72	0.75	0.70	0.66
		untrained	0.71	0.34	0.54	0.77	0.71	0.69	0.71	
	Spec2VolCAMU-Net	trained	0.69	0.46	0.57	0.63	0.66	0.67	0.69	0.68
		untrained	0.74	0.45	0.63	0.75	0.76	0.68	0.70	
	NT-ViT	trained	0.67	0.68	0.69	0.67	0.67	0.51	0.53	0.62
		untrained	0.69	0.73	0.71	0.72	0.70	0.65	0.63	
	Band power / information-matched		0.79 / 0.83					0.80 / 0.83	0.81 / 0.80	

Cohort	Synthesizer	Weights	$k = 20$	$k = 5$	$k = 10$	$k = 40$	Blocks	Covariates	ICA	XP1
	Band power without 30–45 Hz; CNN-TAG 50 epochs		0.78							0.65
Fribourg	CNN-TAG	trained	0.60	0.49	0.55	0.63	0.65	0.58	0.57	0.37
		untrained	0.55	0.44	0.43	0.60	0.68	0.62	0.54	
	E2fNet	trained	0.54	0.58	0.51	0.51	0.56	0.56	0.54	0.54
		untrained	0.54	0.30	0.38	0.51	0.63	0.51	0.50	
	Spec2VolCAMU-Net	trained	0.61	0.52	0.63	0.61	0.60	0.62	0.62	0.52
		untrained	0.49	0.46	0.39	0.57	0.45	0.55	0.52	
	NT-ViT	trained	0.63	0.63	0.63	0.63	0.67	0.48	0.37	0.34
		untrained	0.53	0.56	0.53	0.52	0.51	0.54	0.36	
	Band power / information-matched		0.54 / 0.56					0.52 / 0.49	0.60 / 0.65	
	Band power without 30–45 Hz; CNN-TAG 50 epochs		0.56							0.66